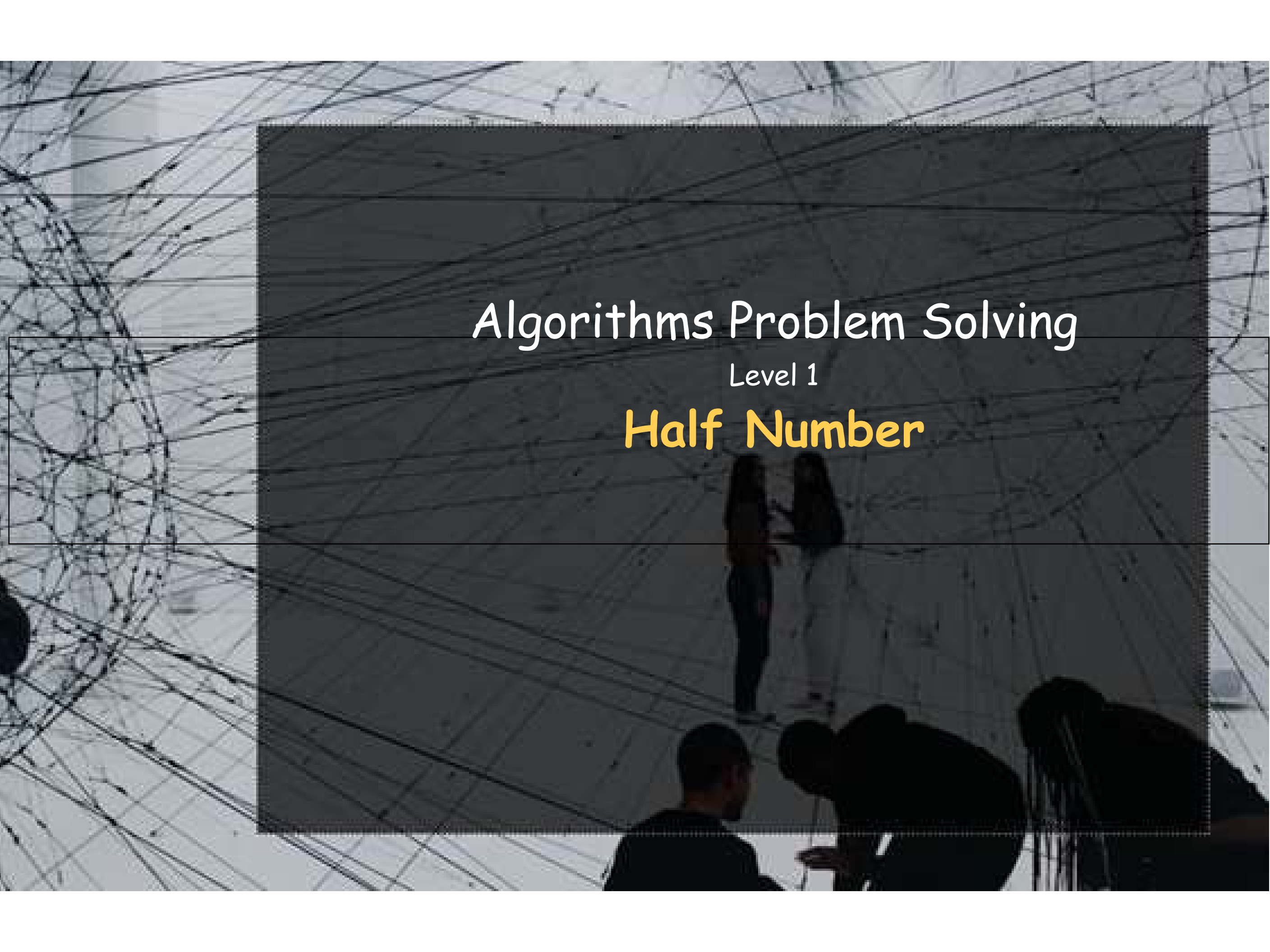


Algorithms Problem Solving

Level 1

Half Number



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Definition

- ◆ This program reads an integer number from the user and calculates half of that number.
- ◆ The program uses separate functions to read the number, calculate its half, and print the result.
- ◆ The result is calculated as a floating-point value so that decimal results are preserved.

Explanation

- ◆ `readNumber()` reads an integer from the user and returns it.
- ◆ `calculateHalfNumber()` receives the integer and divides it by 2.
- ◆ The expression `(float)Num / 2` converts `Num` to float before division.
- ◆ Because the function returns float, results such as 5.5 can be represented correctly.
- ◆ `printNumber()` prints the original number and the calculated half.
- ◆ `to_string()` converts the numeric values into strings for output.
- ◆ In `main()`, the functions are combined:
`printNumber(readNumber());`

Steps in main()

1. Read an integer number from the user.
2. Return the number from readNumber().
3. Pass the number to printNumber().
4. Call calculateHalfNumber() to calculate half of the number.
5. Convert the values to strings using to_string().
6. Print the original number and its half.

💡 Example

◆ If the user enters:

11

◆ calculateHalveNumber(11) calculates:

$11 / 2 = 5.5$

➡ Result

The program prints:

Half of 11 is: 5.500000

Note

- ◆ `(float)Num` performs an explicit type conversion from `int` to `float`.
- ◆ This allows decimal results instead of integer division.
- ◆ The function name `claculateHalveNumber` contains a spelling mistake; a clearer name would be `calculateHalfNumber`.

⊘ Warning

- ◆ If Num were divided by 2 as an integer without converting it to float, the decimal part would be lost.
- ◆ For example, $11 / 2$ using integer division gives 5, while $(\text{float})11 / 2$ gives 5.5.



Comparison

▨ Integer Division

- ✓ Both operands are integers.
- ✓ The decimal part is removed.
- ✓ Example: $11 / 2 \rightarrow 5$.

▤ Floating-Point Division

- ✓ At least one operand is a floating-point value.
- ✓ The decimal part is preserved.
- ✓ Example: $(\text{float})11 / 2 \rightarrow 5.5$.

✦✦ Summary

- ✓ The program separates input, calculation, and output into different functions.
- ✓ Casting an integer to float preserves the decimal part when dividing by 2.
- ✓ `to_string()` converts numeric values into strings for output.

Thank you 😊